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**On a General Means
of Verifying the Expression for the Potential
Relative to an Arbitrary Mass,
Homogeneous or Heterogeneous.**

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**On a General Means of Verifying the Expression
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Given an arbitrary mass, bounded in every direction, forming a continuous whole or composed of several separated parts, and having at each of its points a finite density, if one forms the sum of all the elements of this mass, each divided by its distance from an arbitrary point m , one obtains what geometers have agreed to call the potential of the mass with respect to that point. It is known that the triple integral thus defined, which is always a determinate function of the rectangular coordinates x, y, z of the point m , has a great number of important properties. I shall recall here only those on which the observation that is the object of this note bears.

1) The potential v and its first-order differential coefficients

$$\frac{\partial v}{\partial x}, \quad \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial z},$$

which express the components of the attraction exerted by the mass at the point m , are finite and continuous functions of x, y, z throughout all of space.

2) There always exist determinate bounds which the products

$$xv, \quad yv, \quad zv, \quad x^2 \frac{\partial v}{\partial x}, \quad y^2 \frac{\partial v}{\partial y}, \quad z^2 \frac{\partial v}{\partial z}$$

cannot exceed at any point of space.

3) If one leaves aside the particular points around which the density varies abruptly, each of the three derivatives

$$\frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 v}{\partial z^2}$$

will always have a finite and unique value, and these simultaneous values will be

such that

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = -4\pi \varrho, \tag{a}$$

where ϱ denotes the density at the point (x, y, z) , to be considered zero when this point lies outside the mass. For the points just excepted, which will always be situated on certain surfaces, among which are also included those which bound the mass, or at least the parts of the latter where the density is not zero, the functions

$$\frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 v}{\partial z^2}$$

will generally each have two distinct values, but always finite ones; for the first, for example, these will be the two limits of the quotient

$$\frac{\Delta \frac{\partial v}{\partial x}}{\Delta x},$$

corresponding to an infinitely small positive or negative value of Δx . From the combination of these double values there will in general result eight different values for the trinomial (a); it is not necessary to consider them. It suffices for our object to know that they are all finite.

Here is now the very simple observation to which the properties just stated give rise, and which, so far as I know, had not yet been made. These properties completely characterize the potential v , and there exists no other function v' possessing all these properties together.

To prove this, let us suppose that the contrary takes place, and see what will result for the difference $v' - v = u$. From the nature of the properties in question, it is evident that the new function u will still satisfy conditions (1) and (2), while equation (a) will be replaced by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

The points mentioned above must, however, be excepted; we know only that finite values of the trinomial correspond to such points. These finite values - unknown for the moment, but in fact all equal to zero, since we are going to see that $u = 0$ - occurring only along certain

surfaces, will have no influence on the result of a triple integration, and we shall have rigorously

$$\iiint u \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz = 0,$$

the limits being arbitrary. Let us extend each of the three integrations from $-a$ to $+a$, and consider in particular the integral of the first term. If we begin with the operation relative to x , integration by parts gives

$$\int_{-a}^{+a} u \frac{\partial^2 u}{\partial x^2} dx = \left(u \frac{\partial u}{\partial x} \right)_{-a}^{+a} - \int_{-a}^{+a} \left(\frac{\partial u}{\partial x} \right)^2 dx,$$

the indices appended to the first term indicating that one must take the difference of the two values corresponding to $+a$ and to $-a$. It is indeed to this difference alone that the terms produced outside the sign by the integration by parts reduce, $u \frac{\partial u}{\partial x}$ being a continuous function of x . Proceeding in an analogous manner on the other two integrals, one obtains

$$\begin{aligned} & \iiint \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 \right) dx dy dz \\ &= \iint \left(u \frac{\partial u}{\partial x} \right)_{-a}^{+a} dy dz + \iint \left(u \frac{\partial u}{\partial y} \right)_{-a}^{+a} dx dz + \iint \left(u \frac{\partial u}{\partial z} \right)_{-a}^{+a} dx dy. \end{aligned}$$

By virtue of conditions (2), the function of y and z that stands under the sign in the first double integral will, throughout the extent of that integral, be numerically less than $\frac{k}{a^3}$, where k does not depend on a ; the integral is therefore less than $\frac{4k}{a}$, and vanishes for $a = \infty$. The same thing holding for the other two, it follows that the triple integral, taken between infinite limits, is likewise zero. But the function subjected to the triple integration, being continuous and never capable of becoming negative, must be identically zero; consequently u has a constant value, which can only be zero, since u must vanish at infinity. This was to be shown.

The result just obtained supplies a general means of verifying the expression for the potential when this potential is given throughout all of space. It will suffice, for this purpose, to make sure that the given expression satisfies all the conditions enumerated above.

The application of this procedure to the known formulae relating to a homogeneous ellipsoid perhaps offers the most expeditious means of establishing the theorems concerning that celebrated question, provided, however, that one seeks only a simple verification and does not have in view a method that naturally leads to the supposedly unknown results at the same time as it proves them.

The equation of the ellipsoid being

$$\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} + \frac{z^2}{\gamma^2} = 1,$$

the trinomial forming its left-hand side will be less than or greater than unity according as the point (x, y, z) is inside or outside the ellipsoid. In the second case, the equation

$$\frac{x^2}{\alpha^2 + \sigma} + \frac{y^2}{\beta^2 + \sigma} + \frac{z^2}{\gamma^2 + \sigma} = 1$$

has a unique positive root σ , which varies continuously with the variables x, y and z . This is seen by observing that the derivatives

$$\frac{\partial \sigma}{\partial x}, \quad \frac{\partial \sigma}{\partial y}, \quad \frac{\partial \sigma}{\partial z},$$

given by the equations

$$l \frac{\partial \sigma}{\partial x} = \frac{2x}{\alpha^2 + \sigma}, \quad l \frac{\partial \sigma}{\partial y} = \frac{2y}{\beta^2 + \sigma}, \quad l \frac{\partial \sigma}{\partial z} = \frac{2z}{\gamma^2 + \sigma}, \quad (b)$$

where

$$l = \frac{x^2}{(\alpha^2 + \sigma)^2} + \frac{y^2}{(\beta^2 + \sigma)^2} + \frac{z^2}{(\gamma^2 + \sigma)^2},$$

always have finite values. This being set, if for brevity we put

$$D = \sqrt{\left(1 + \frac{s}{\alpha^2}\right) \left(1 + \frac{s}{\beta^2}\right) \left(1 + \frac{s}{\gamma^2}\right)},$$

it remains to prove that one has

$$v = \pi \int_0^\infty \left(1 - \frac{x^2}{\alpha^2 + s} - \frac{y^2}{\beta^2 + s} - \frac{z^2}{\gamma^2 + s}\right) \frac{ds}{D},$$

or

$$v = \pi \int_\sigma^\infty \left(1 - \frac{x^2}{\alpha^2 + s} - \frac{y^2}{\beta^2 + s} - \frac{z^2}{\gamma^2 + s}\right) \frac{ds}{D},$$

according as the point (x, y, z) is located inside or outside the mass, whose density is assumed equal to unity. Differentiation gives

$$\frac{\partial v}{\partial x} = -2\pi x \int_0^\infty \frac{ds}{(\alpha^2 + s)D},$$

or

$$\frac{\partial v}{\partial x} = -2\pi x \int_\sigma^\infty \frac{ds}{(\alpha^2 + s)D},$$

the term arising from the variability of the limit σ in the second case vanishing by virtue of the equation of which σ is a root. It is manifest that the first expressions for v and $\frac{\partial v}{\partial x}$, and the analogous expressions for $\frac{\partial v}{\partial y}$, $\frac{\partial v}{\partial z}$, vary continuously with the coordinates x, y, z , and that the

same is true of those corresponding to an exterior point; one also sees that continuity is not broken at the surface where $\sigma = 0$, which makes the corresponding expressions coincide.

Conditions (2) plainly hold for an interior point and, in this case, are already included in conditions (1). To verify them outside the ellipsoid, let λ be the smallest of the semiaxes α, β, γ . Observing that the factor $\frac{ds}{D}$ in the second expression for v is less than unity, and attending only to numerical values, we evidently have

$$v < \frac{2\pi\alpha\beta\gamma}{(\lambda^2 + \sigma)^{1/2}}, \quad \frac{1}{x} \cdot \frac{\partial v}{\partial x} < \frac{4\pi\alpha\beta\gamma}{3(\lambda^2 + \sigma)^{3/2}}.$$

As, on the other hand, the equation in σ gives, apart from the sign, $x < (\alpha^2 + \sigma)^{1/2}$, one concludes

$$xv < 2\pi\alpha\beta\gamma \left(\frac{\alpha^2 + \sigma}{\lambda^2 + \sigma} \right)^{1/2}, \quad x^2 \frac{\partial v}{\partial x} < \frac{4}{3}\pi\alpha\beta\gamma \left(\frac{\alpha^2 + \sigma}{\lambda^2 + \sigma} \right)^{3/2},$$

inequalities whose right-hand sides remain finite as σ increases from zero to infinity.

Let us pass to the derivatives of the second order. One finds

$$\frac{\partial^2 v}{\partial x^2} = -2\pi \int_0^\infty \frac{ds}{(\alpha^2 + s)D}$$

or

$$\frac{\partial^2 v}{\partial x^2} = \frac{2\pi x}{(\alpha^2 + \sigma)\Delta} \cdot \frac{\partial \sigma}{\partial x} - 2\pi \int_\sigma^\infty \frac{ds}{(\alpha^2 + s)D},$$

where Δ denotes the value of D corresponding to $s = \sigma$. These expressions are still finite and determinate, but they no longer coincide at the surface. If we now add to each of these equations the two analogous equations, and observe that one has, indefinitely,

$$\int \left(\frac{1}{\alpha^2 + s} + \frac{1}{\beta^2 + s} + \frac{1}{\gamma^2 + s} \right) \frac{ds}{D} = -\frac{2}{D} + \text{const.},$$

there will come, according to the two cases,

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = -4\pi,$$

or

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = \left(\frac{2x}{\alpha^2 + \sigma} \cdot \frac{\partial \sigma}{\partial x} + \frac{2y}{\beta^2 + \sigma} \cdot \frac{\partial \sigma}{\partial y} + \frac{2z}{\gamma^2 + \sigma} \cdot \frac{\partial \sigma}{\partial z} - 4 \right) \frac{\pi}{\Delta}.$$

The first of these two equations already agrees with condition (a), and to make sure that the same is true of the second it suffices to take the sum of the three equations (b), each multiplied by its right-hand side, and then to suppress the common factor l , which is evidently different from zero. One thus recognizes that the right-hand side of the last equation vanishes, as it must for an exterior point.

We have supposed up to now that the given mass has three dimensions. When this mass has only two and is distributed over one or over several surfaces, the method of verification just set out will have to undergo certain modifications, which nevertheless follow too simply from the known theorems relating to that case for it to be necessary to enter into further details on this point.¹

¹Cf. the immediately following paper. K.